Paleomagnetism of Siluro-Devonian and Cambrian granitic rocks from the Avalon zone in Cape Breton Island, Nova Scotia

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As part of a major study of the Avalon zone in Cape Breton Island, we describe the paleomagnetism of radiometrically well-studied granitic rocks of two generations, representing the Acadian (350–450 Ma) and the Avalonian (520–580 Ma) Orogenies, respectively. Detailed alternating field (AF) and thermal experiments performed on the younger granites indicate that they are characterized by three different mean directions of magnetization in the 10–60 mT coercivity spectra: SE ($D = 149°$, $I = +70°$; $K = 160$, $N = 4$ sites); NE-1 ($D = 72°$, $I = -70°$; $K = 320$, $N = 3$ sites); and NE-2 ($D = 31°$, $I = -24°$; $K = 67$, $N = 3$ sites) with corresponding paleopole positions at CB-1 (16°N, 41°W; δp, δm = 6°, 7°); CB-2 (32°N, 97°W; δp, δm = 7°, 8°); and CB-3 (27°S, 96°W; δp, δm = 6°, 11°). The radiometric ages of these three remanence directions do not differ from each other by more than 50 Ma; these remanences are also present as low coercivity (5–25 mT) magnetizations in the Avalonian granites (10 sites). Antipole CB-3 agrees with other recently reported Early Devonian paleomagnetic results from the northern Appalachians. In contrast, the two other antipoles (CB-1 and CB-2) are located in southerly latitudes but do not differ significantly from poles of contemporaneous rocks in New Brunswick and Newfoundland. The "aberrancy" of southerly located Siluro-Devonian poles with respect to those in northern latitudes is discussed in the context of possible transcurrent motion of the "Acadia" displaced terrain with respect to cratonic North America. Presently available paleomagnetic data are not conclusive and cannot confirm or negate this possibility. The CB-1, CB-2, and CB-3 poles are interpreted as representing rapid apparent polar wander with respect to Cape Breton Island during Siluro-Devonian time.

In the older Cambrian granites, pole CB (37°N, 176°E; δp, δm = 2°, 3°), derived from a high coercivity (20–70 mT) mean remanence direction NW ($D = 318°$, $I = +15°$; $K = 298$, $N = 10$ sites), probably corresponds to the Avalonian Orogeny. Further results are needed to interpret the paleogeographic setting of the Avalon microcontinent in early Paleozoic time.

Nous décrivons le paléomagnétisme de roches granitiques de deux générations, ceci faisant partie d'une étude majeure de la zone d'Avalon de l'Ile du Cap Breton. Il s'agit de roches granitiques dont la radiométrie a été largement étudiée et qui représentent respectivement les orogénies acadienne (350–450 Ma) et avalonnienne (520–580 Ma). Des expériences FA et thermiques réalisées sur des granites plus récents indiquent qu'ils sont caractérisés par trois directions moyennes différentes de magnétisation dans le spectre de coercivité: SE ($D = 149°$, $I = +70°$; $K = 160$, $N = 4$ sites); NE-1 ($D = 72°$, $I = 70°$; $K = 320$, $N = 3$ sites); NE-2 ($D = 31°$, $I = -24°$; $K = 67$, $N = 3$ sites) avec des positions correspondantes des paléopôles à CB-1 (16°N, 41°O; δp, δm = 6°, 7°); CB-2 (32°N, 97°O; δp, δm = 7°, 8°) et CB-3 (27°S, 96°O; δp, δm = 6°, 11°). Les âges radiométriques de ces trois directions de remanence ne diffèrent entre elles par pas plus de 50 Ma; ces rémanences sont également présentes comme faibles magnétisations de coercivité (5–25 mT) dans les granites avalonniens (10 sites). L'antipôle CB-3 concorde avec d'autres résultats paléomagnétiques récemment rapportés du Dévonien inférieur des Appalaches du Nord. Opposément, les deux autres antipôles (CB-1 et CB-2) sont localisés aux latitudes sud mais ne diffèrent pas de manière significative du pôle des roches contemporaines du Nouveau-Brunswick et de Terre-Neuve. La localisation "aberrante" des pôles du Siluro-Dévonien au sud par rapport à ceux des latitudes nord est interprétée par un contexte possible de mouvement transcourant du territoire déplacé de l'Acadie par rapport au craton de l'Amérique du Nord. Les données paléomagnétiques actuellement disponibles ne permettent pas de conclure et ne peuvent ni confirmer ou infirmer cette hypothèse. Les pôles CB-1, CB-2 et CB-3 sont interprétés comme représentant une dérivation rapide du pôle apparent par rapport à l'Ile du Cap Breton durant l'âge Siluro-Dévonien.

Dans les granites plus anciens d'âge Cambrien le pôle CB (37°N, 176°E; δp, δm = 2°, 3°) dérivé à partir d'une forte coercivité (20–70 mT) avec direction moyenne de rémanence NO ($D = 318°$, $I = +15°$; $K = 298$, $N = 10$ sites), correspond probablement à l'orogénie avalonnienne. Des recherches supplémentaires sont nécessaires pour interpréter la position paléogéographique du microcontinent d'Avalon du Paléozoïque inférieur.

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Introduction

Avalon zone—its extent and tectonic evolution

Williams (1978*a*, 1979) proposed a five-fold division of the Appalachian Orogen applicable to its entire length. The Avalon zone forms one of the two easterly zones, which is wider than the combined width of the other zones (Lilly 1965; Pelletier 1971; Poole 1976; Haworth and Lefort 1979; Haworth and Keen 1979;

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Haworth 1980) and is underlain predominantly by Upper Precambrian volcanic and sedimentary rocks with minor unconformities. The type of deformation is generally simple to moderate and metamorphism is low greenschist to subgreenschist facies in the Upper Precambrian rocks, which are locally cut by Precambrian–Cambrian granites; elsewhere the succession passes upwards conformably into lower Paleozoic shales and sandstones containing Acado-Baltic faunas. These rocks are generally referred to as "Avalonian" after the type locality on the Avalon Peninsula, e.g., "Avalon Platform" (Kay and Colbert 1965), "Avalon geosyncline" (Poole 1967, 1976), and "Avaland microcontinent" (Williams 1978*a*). The geology of the Avalon zone has been amply reviewed in the recent literature (Williams *et al.* 1972, 1974; Rodgers 1972; King *et al.* 1974; Rast *et al.* 1976; Schenk 1978; Poole 1976; Strong *et al.* 1978; Strong 1979*a*; O'Driscoll and Strong 1979; King 1979; Haworth 1980). Similar Avalonian type Upper Precambrian volcanic and intrusive rocks overlain by Cambro-Ordovician strata are found on Cape Breton Island, in eastern New Brunswick, Massachusetts, and farther south in the Carolinas. Keppie (1978) refers to Avalonian rocks in the Antigonish Highlands and Cobequid Mountains of Nova Scotia. Presumed correlatives of the Avalon zone occur also in Wales, Brittany, northwest Africa, the Iberian Peninsula, Czechoslovakia, and Turkey (Williams 1978*b*, 1979).

Despite the well-known geology of the Avalon zone at different localities, the controlling tectonic feature responsible for its Precambrian development is still poorly understood. The two major hypotheses on the Avalon zone postulate, respectively, an archipelago of volcanic islands (Hughes 1972; Hughes and Brückner 1972; Rast *et al.* 1976) and a "Basin and Range" type rifting environment (Papezik 1970, 1972), with or without associated subduction (Strong *et al.* 1974; Strong and Minatides 1975; Rankin 1975, 1976; Giles and Ruitenberg 1977; Strong *et al.* 1978).

While the age, duration, and significance of the deformational history of the Avalon zone are not well known, there appears to be a general agreement that it was mainly affected by two major orogenic events: (1) the Late Precambrian Avalonian Orogeny (Lilly 1966; Hughes 1970; Kennedy 1976), which is nearly contemporaneous with Cadomian and Pan-African orogenic events of Europe and Africa (Cogné and Wright 1980); and (2) the apparently more intense Devonian Acadian Orogeny. Present geological knowledge of either the tectonic settings or the time and duration of deformational events is too scanty to permit an adequate understanding of the evolution of the Avalon zone. Van der Voo (1979) speculates that the Avalon zone forms part of a post-Cambrian to pre-Devonian "Armorica" plate because of EoCambrian paleomagnetic results based on the Armorican massif (France), the Bohemian massif (Czechoslovakia), the Baltic Shield – Russian platform, and southern England and Wales.

In such a context paleomagnetism can be used as a basic tool to determine the details of block movements and the extent of postulated Hadrynian–Paleozoic oceans. It can help to constrain different tectonic models by providing paleolatitudes and relative orientations of adjacent continental terranes.

Previous paleomagnetic studies on the Avalon zone

The first paleomagnetic study on the Avalon rocks was by Nairn *et al.* (1959) on the upper Hadrynian Signal Hill and Blackhead red sandstones of the Cabot Group in eastern Newfoundland. The result was based only on the natural remanent magnetization (NRM) of the sediments, as cleaning techniques were unavailable at that time (pole SH, Table 6). Black (1964) studied the Lower Cambrian Ratcliffe Brook red sandstones of New Brunswick using alternating field (AF) demagnetization only up to 30 peak mT (pole RB, Table 6), which may be insufficient to erase any secondary components of relatively high coercivity; he did not employ stepwise thermal demagnetization, which is generally a much more effective treatment for obtaining a reliable paleomagnetic signal in red beds. The authors explained the discordance of the paleomagnetic poles with respect to contemporaneous poles from mainland North America in terms of Wegener's (1929) hypothesis of rotation of Newfoundland. The results of these two investigations, the Cabot Group and Ratcliffe Brook data, were later disputed on different grounds by Du Bois (1959) and Robertson *et al.* (1968), respectively. Deutsch (1969) argued that the results could equally well be interpreted as supporting Wilson's (1966) model for a proto-Atlantic ocean since the Ratcliffe Brook sampling sites in New Brunswick were part of the Avalon tectonostratigraphic zone. A detailed paleomagnetic study of Lower Ordovician Wabana red sandstones from eastern Newfoundland (Deutsch and Rao 1970) resulted in an apparently discordant paleomagnetic pole (W, Table 6). Extensive paleomagnetic investigations of Avalonian rocks from the Atlantic provinces, ranging in age from Hadrynian to Cambrian, have been in progress since 1977 at Université Laval, Memorial University of Newfoundland, and the Earth Physics Branch, Ottawa.

As a part of that program, the present paleomagnetic investigation is the first report of a study area chosen in Cape Breton Island (Fig. 1), which is apparently the only area in the Canadian Appalachians where large bodies of granitic intrusives of pre-Early Cambrian age (Cormier 1972) are prominent along with the Siluro-Devonian granitic stocks exposed on a regional scale and where most of the granitic intrusives are isotopically well studied (Fairbairn *et al.* 1960; Wanless *et al.* 1968;

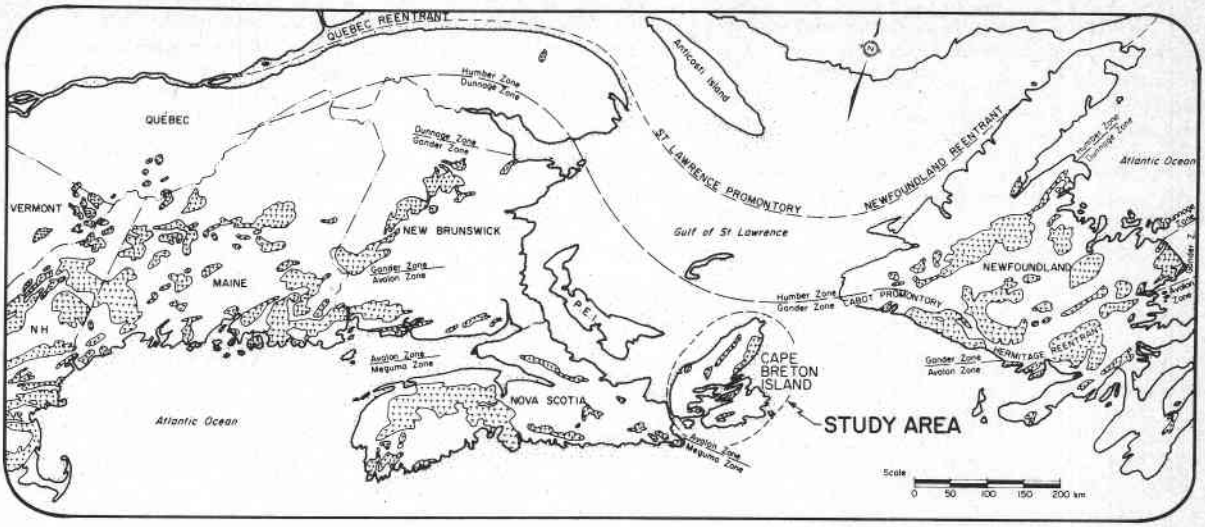


FIG. 1. Distribution of Paleozoic (mainly Devonian) granitic plutons in northern Appalachians (after Williams 1978*a*). The study area is indicated.

Wanless 1969; Cormier 1972; Keppie and Smith 1978; Cormier 1979).

Geologic setting and age of granitic rocks

Granitic and associated Helikian to Carboniferous rocks are widespread in the northern Appalachians but the bulk apparently are of Siluro-Devonian age, a time span that occupies more than 25% of the exposed Appalachian Orogen in Canada, with granites constituting by far the most voluminous single rock type (Fig. 1). Cambrian granites with isotopic ages around 500–600 Ma occur in the Avalon zone of southeastern Newfoundland (McCartney *et al.* 1966; Pelletier 1971; Frith and Poole 1972; Bell *et al.* 1977), central and southern parts of Cape Breton Island (Fairbairn *et al.* 1960; Wanless *et al.* 1968; Cormier 1972; Keppie and Smith 1978), northern Nova Scotia (Benson 1974), and southern New Brunswick (Leech *et al.* 1963). Geochronological studies of Siluro-Devonian granites that locally cut across the Cambro-Ordovician tectonostratigraphic boundaries reveal a wide range of isotopic ages between 450 and 350 Ma, those that occur in the Avalon zone are exposed mostly on northern Cape Breton Island (Cormier 1972; Wiebe 1972, 1975) and in southwestern New Brunswick and adjacent Maine (Pajari *et al.* 1974).

Granitoid plutonism in Cape Breton Island is abundant; the distribution and geology of the plutons covered in the present study are described in detail by Barr *et al.* (1978). In the earlier geological literature (Hayes and Bell 1923; Bell and Goranson 1938; Weeks 1947, 1954; Hutchinson 1952) it is widely presumed that most of the granitic intrusions belong to the post-Cambrian to pre-Carboniferous age interval based mainly on conjecture without any specific reference to field relationships between granitic intrusives and adjacent country rocks. Rb–Sr and K–Ar mineral ages on granitic plutons from different localities on the island by Fairbairn *et al.* (1960) and Wanless *et al.* (1968) and later whole-rock Rb–Sr age measurements by Wanless (1969) have revealed that their ages range variably from 354 to 587 Ma. Using a value of 1.39×10^{-11} year^{-1} for the decay constant, numerous Rb–Sr whole-rock and mineral age determinations were done on Cape Breton granites and reported later by Cormier (1972), who indicated for the first time that most of the granites have primary ages closer to 560 Ma, whereas those in the northern part have considerably younger Siluro-Devonian primary ages, 350–450 Ma. Following the standard set of decay constants and isotopic abundances of Steiger and Jäger (1977), all the previously determined isochrons were recalculated and compiled by Keppie and Smith (1978).

Cormier's (1972) isotopic dates on granites from northern Cape Breton are supported by field studies (Wiebe 1972, 1973, 1975) indicating that the history of these granitic rocks is not as simple as that of granites from the southeastern part of the island. In order to explain the younger ages of granites in his data, Cormier (1972) suggested two interpretations. One is that the granitic rocks of the northern Highlands were all emplaced at about the same time as those elsewhere on the island, around 560 Ma ago, but the former were later affected by the Acadian Orogeny, which led to a complete updating of biotites in many instances. The other interpretation is that there exist at least two generations of Cambrian and Devonian granites as suggested by Wiebe (1972) in his structural studies. Cormier (1973) also noted that, since the older granitic

TABLE 1. Location of paleomagnetic sampling sites (Fig. 2) and the corresponding isotopically dated localities*

Paleomagnetic sampling site	Type of granitic rock	No. of the nearby isotopically dated locality	Location	Whole-rock/ mineral isochron (Ma)	Reference
GA-1, -2	Syenodiorite	6	Coxheath Hills	584 ± 28	Wanless *et al.* (1968)
				580 ± 70	Cormier (1972)
GA-3, -4	Granodiorite	5	Boisdale Hills	493	Wanless *et al.* (1968)
				519 ± 70	Cormier (1972)
GA-5	Syenite	4	George River Station	564 ± 70	Cormier (1972)
GA-6	Pink leucogranite	3	Long Island Quarry	576 ± 20	Cormier (1972)
				587 ± 10	Wanless (1969)
GA-7, -8	Pink leucogranite	7	Kelly Mountain	568 ± 35	Cormier (1972)
GA-9	Pink leucogranite		St Ann's Mountain (Barr *et al.* 1978)	—	—
GA-10	Porphyritic granite	9	Craignish Hills	552 ± 60	Cormier (1972)
GB-1, -2	Pink biotite granite	13	Cape Smoky pluton (Barr *et al.* 1978)	460 ± 60	Cormier (1972)
GB-3, -4	Pink biotite granite	13	Cape Smoky pluton (Barr *et al.* 1978)	460 ± 60	Cormier (1972)
GB-5	Pink biotite granite		8 km south of Black Brook Cove on Cabot Trail	—	—
GB-6	Biotite granite gneiss, coastal gneisses (Wiebe 1972)	15	2 km north of Neil's Harbour	562 ± 80	Cormier (1972)
GB-7	Muscovite biotite granite	14	12 km north of Neil's Harbour, White Point pluton (Barr *et al.* 1978)	410 ± 50	Cormier (1972)
GB-8	Biotite granite	15	1 km south of Black Brook Cove	562 ± 80	Cormier (1972)
				538	Cormier (1972)
				373	Cormier (1972)

*See index map, Fig. 1 in Cormier (1972).

rocks predate the Lower Cambrian sedimentary rocks that overlie them in southeastern Cape Breton Island, the granites can be considered to be pre-Early Cambrian or Eocambrian. Such an age span would correspond to the intrusive activity in the Avalonian Orogeny if one were to follow the interpretation by Rodgers (1970) who associated the time of intrusion of the Holyrood granite (607 ± 11 Ma: McCartney *et al.* 1966) in eastern Newfoundland and the Hoppin Hill granite (570 ± 20 Ma: Fairbairn *et al.* 1967) in southeastern Massachusetts with the Late Precambrian Avalonian Orogeny, which affected the southeastern continental margin of the presumed Proto-Atlantic ocean. Paleomagnetic studies on such well-studied Upper Precambrian to Cambrian and Siluro-Devonian granitic intrusions associated, respectively, with the Avalonian and the Acadian orogenic events, based on their time of emplacement and deformation, impose important constraints on the possible tectonic history of the Avalon zone.

Paleomagnetic sampling and measurements

In the present paleomagnetic study, 100 oriented hand samples at 18 sites (with a minimum of five samples per site) were collected from radiometrically well-studied localities (index map Fig. 1 in Cormier 1972) in the central and northern parts of Cape Breton Island. Based on radiometric data, the collection was concentrated at 10 sites on the plutons that yielded primary ages of 520–580 Ma in the central part. These were arbitrarily denoted as "GA" sites to distinguish them from the other eight "GB" sites, which come mainly from the Cape Smoky and White Point plutons (Barr *et al.* 1978) of the northern Highlands and are mostly characterized by younger ages (between 350 and 450 Ma), except sites 6 and 8 (Table 1; Fig. 2).

The granitic rocks from the GA and GB sites are mostly fine- to medium-grained and massive, and apparently unfoliated and unmetamorphosed. The type of granitic rock collected from each sampling site is listed in Table 1. At all the sampling sites, contact relations of these rocks with younger Carboniferous sedimentary rocks are nonconformable.

Magnetization (M) was measured with a digital Schonstedt spinner magnetometer (DSM-1); the volume susceptibility (K) was measured at room temperature in the Earth's field ($H = 0.05$ mT), using a Cenco MS-3B susceptibility bridge. Two methods of demagnetization were carried out using (1) an AF demagnetizer with three-axis tumbler operating in residual fields of about 5 nT (Seguin 1975) and (2) a Schonstedt thermal demagnetizer (TSD-1) with residual fields of less than 15 nT. Stepwise AF and thermal procedures were employed up to 100 mT and 700°C, respectively. For the NRM of each oriented sample at least two specimens (diameter 2.5 cm, height 2.2 cm) were used to obtain a

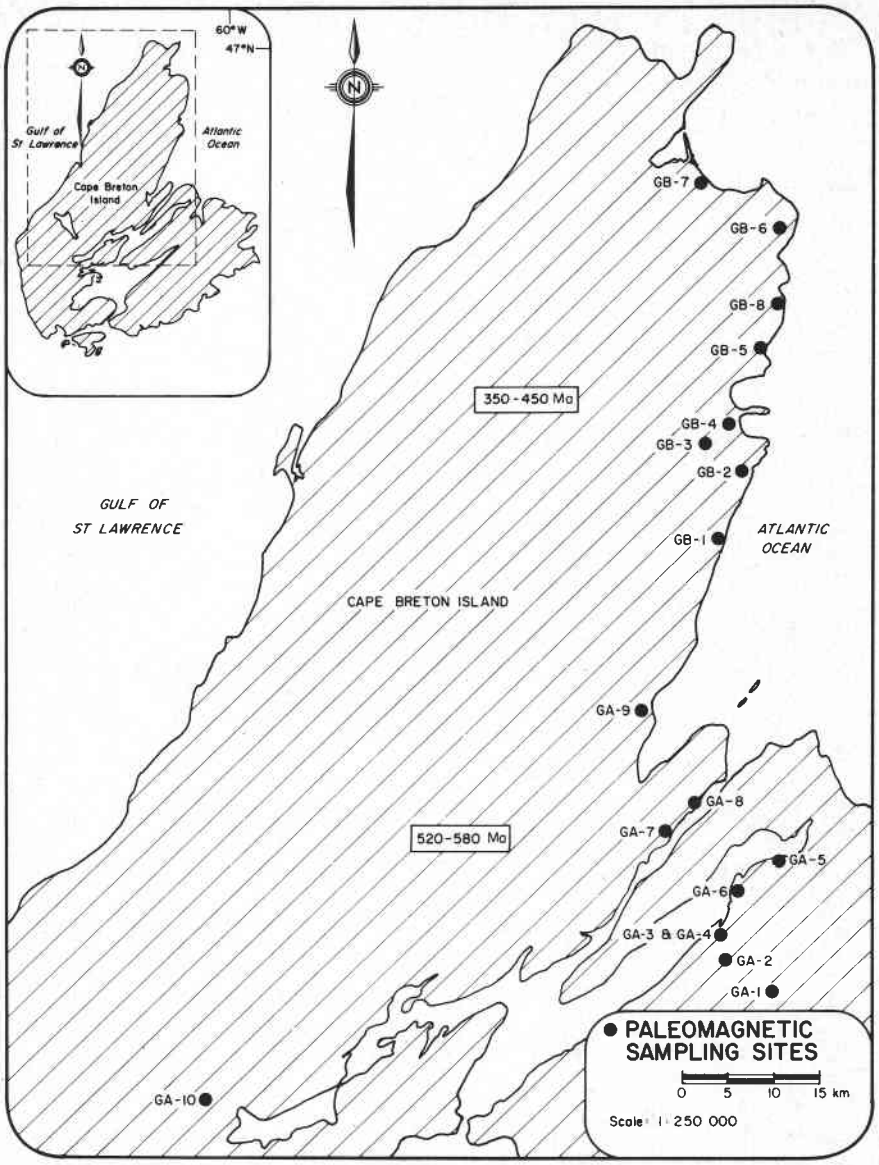


FIG. 2. Distribution of paleomagnetic sampling sites (GA and GB) in Cape Breton Island. For details on the geographic locations of the sites, refer to Table 1 and Fig. 1 (index map) of Cormier (1972) and Table 1 for the corresponding isotopic data.

sample mean direction. Single measurements of NRM were usually repeatable to within 3% in intensity (J) and within 2° in direction (declination D, inclination I); the total estimated uncertainty in the NRM determinations (combined orientation and measurement error) was about 5% in intensity and 4° in direction.

NRM of GB and GA granites

GB collection

Individual specimen NRM directions of GB granites (Fig. 3), although characterized by wide scatter as a result of positively and negatively inclined steep and shallow remanences, have a pronounced deviation away from the present Earth's field (PEF) ($D = 334°$, $I = +71°$) at the site localities, indicating that these granites carry some ancient remanence especially since the directions of the remanence of the few specimens (<10%) that lie close to PEF were found to move away from those initial directions during subsequent AF or thermal treatment. The wide scatter in NRM with dual polarity of remanences is therefore indicative of the presence of both low and high coercivity components. The intensities of NRM (J_n) vary from 4×10^{-4} to 7×10^{-2} A/m with the values of Koenigsberger ratio (Q_n =

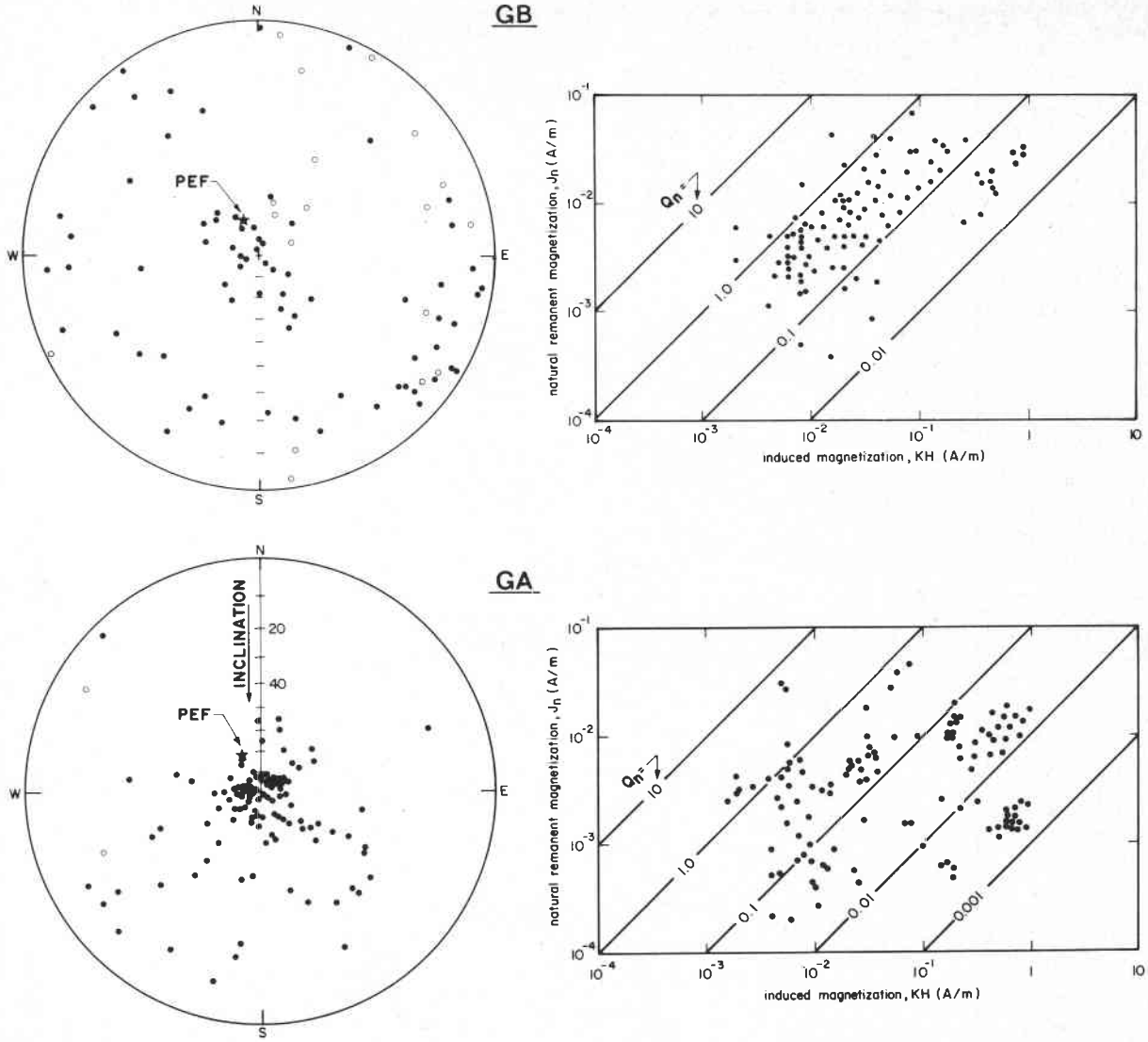


FIG. 3. Individual specimen NRM directions plotted on Wulff net for GB and GA granites (left). Solid (open) circles represent projections on lower (upper) hemisphere. The star represents the present Earth's field (PEF) at the sampling region. Plots of corresponding intensity of NRM (J_n in A/m) against the induced magnetization (KH in A/m) with resulting slope lines indicating the values of Koenigsberger ratio, Q_n (right).

J_n/KH) between 0.01 and 10 but with predominant ratios between 0.1 and 1.0.

GA collection

Unlike the GB collection, the GA granites are less widely scattered in their NRM directions and are characterized by steep positive inclinations in the majority of specimens (>60%) (Fig. 3). Some specimens (≃20%) possess directions close to PEF, but during subsequent AF demagnetization were found to move away from their initial directions at fields as low as 5 mT, indicating the presence of very minor soft components of magnetization along the PEF. The intensities of magnetization of the GA and GB granites occupy approximately the same range, but 20% of the former were found to possess weaker intensities of magnetization (10^{-4}–10^{-3} A/m) than the latter. Unlike in the GB collection, the Q_n ratios in GA rocks span five orders of magnitude (0.001–10) and this wide variation could be due to their earlier age of emplacement and (or) subsequent geological history. The wide range of intensities could also be due to the variable percentage of opaques present in the granitic rock types in the collections (Table 1). During the course of this investi-

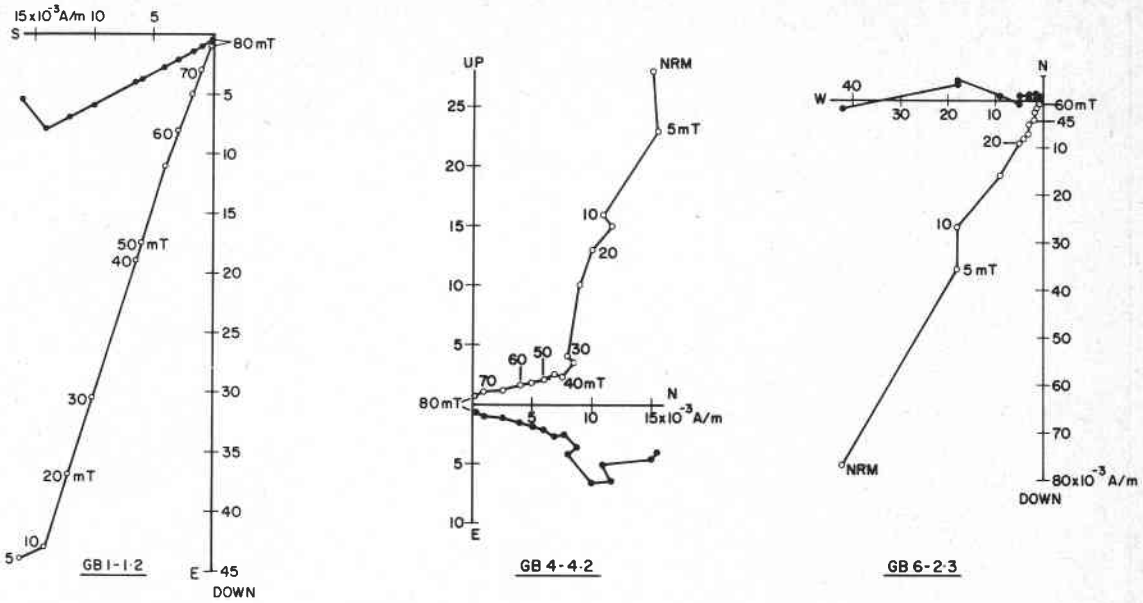


FIG. 4. Representative orthogonal AF demagnetization diagrams (Zijderveld 1967) of successive end points of magnetization vector for three GB granite specimens as indicated. Solid (open) circles represent projections on horizontal (vertical) plane. Numbers next to open circles represent the demagnetization steps (in mT). Units indicated for axes are in 10^{-3} A/m (10^{-6} emu cm^{-3}).

gation (approximately 10 months) the specimens were kept at random in the laboratory in about 0.05 mT field. Repeat measurements of NRM on the majority of the specimens show no significant changes in D, I, or M from the directions obtained initially.

Preliminary demagnetization studies

GB granites

For the preliminary AF study at least three samples per GB site were demagnetized in 15 steps up to 80 mT, and vector diagrams (Zijderveld 1967) were used in the subsequent analysis. For most samples, the intensities of magnetization at or before the 80 mT treatment had generally fallen to less than 5% ($\leq 10^{-3}$ A/m) of the NRM intensity beyond which measurements could not be made due to the acquisition of spurious magnetizations at higher demagnetizing fields. Remanence behaviour during AF treatment typical of most samples from the eight GB sites is shown by three specimens from three different sites (Fig. 4). Vector trajectories are linear until stable end points are reached at higher fields. The linear trajectory for specimen GB 1-1.2 indicates the presence of a stable, single component of magnetization with a southeasterly declination and steep downward inclination. The fact that the decay of the intensity vector is linear from above 10 mT to the origin indicates that, in all likelihood, only one component of magnetization (denoted as SE remanence) is being removed. The SE remanence is found to be the relatively hard component at GB sites 1, 2, and 7 ($H_c \simeq$ 15–60 mT). A soft component of magnetization directed towards the PEF present in some samples is removed by treatment of 5–10 mT. At sites 6 and 8, the SE remanence is found to be a secondary remanence associated with lower coercivities (<30 mT) whereas the hard component is similar to the one observed in the GA granites (NW remanence, described later).

The orthogonal plot for specimen GB 4-4.2 (Fig. 4) shows the presence of two components. One has a northeasterly directed magnetization having a steep to intermediate upward inclination ($D = 65°$, $I = -70°$), which is removed apparently between 30 and 40 mT (denoted as NE-1 remanence). A second, harder remanence (denoted as NE-2) was isolated beyond 40 mT with a linear decay towards the origin; this has a northeasterly direction with a shallow upward inclination ($D = 22°$, $I = -30°$). The NE-2 remanence is present in all site 4 samples with or without the lower coercivity NE-1 remanence whereas at site 3 NE-1 is found to be the only single remanence present between the 5 and 60 mT steps as represented by linear trajectories with a mean direction at $D = 65°$, $I = -70°$. At site 5, the NE-1 remanence is also present but roughly opposite in direction ($D = 267°$, $I = +69°$), together with the higher coercivity NE-2 remanence. At site 6 the NE-2 occurs as the only single remanence in two

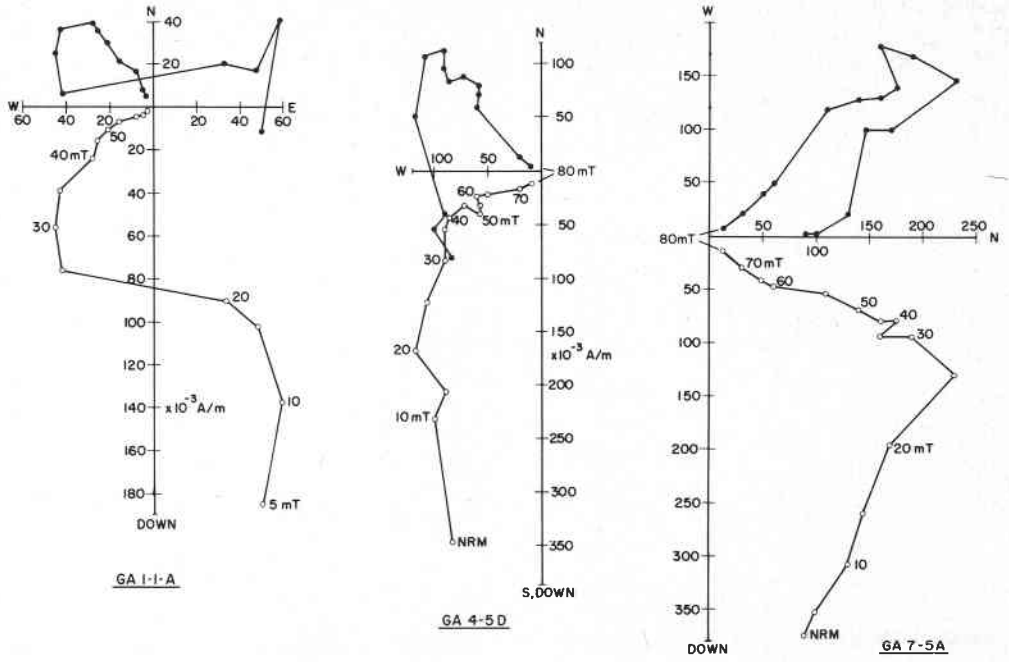


FIG. 5. Representative orthogonal AF demagnetization diagrams (Zijderveld 1967) of successive end points of magnetization vectors for three GA granite specimens as indicated. The other symbols are as in Fig. 4.

samples as shown by specimen GB 6-2.3 (Fig. 4). However, the other three samples at site 6 possess the NE-2 remanence in the low coercivity spectrum along with the higher coercivity NW remanence obtained from the majority of GA granites (see below).

Preliminary thermal demagnetizations of fresh specimens from the samples used in AF treatment were made in order to obtain more information about the SE, NE-1, and NE-2 remanences. The SE remanence was found to be single and stable with unblocking temperatures (T_{UB}) (Roy *et al.* 1979) between 400 and 500°C, suggesting magnetite as the probable carrier of magnetization in GB 1-1.1; thermal demagnetization of specimen GB 6-2.1 revealed the NE-2 remanence as found in GB 6-2.3 with T_{UB} = 300–500°C. Specimen GB 4-4.1 after thermal treatment revealed the NE-2 remanence with a T_{UB} spectrum of 300–500°C. However, the distinction between the T_{UB} spectra was not as sharp as between the coercivity spectra as seen as in its companion specimen during AF treatment.

For the majority of GB specimens, AF treatment gave a better resolution of the different components than did the thermal treatment, although the two methods isolated the same remanence directions. Thermal results were preferred when in some instances a clear decay to the origin could not be obtained beyond 80 mT. In view of the observed number of components (SE, NE-1, and NE-2, the latter sometimes with opposite polarity) and their variability from site to site in the GB granites, and to ascertain the reliability of these components, detailed AF demagnetizations were made on more than 60 specimens (≃70%). Vector diagrams were used in each case to recognize and analyse the respective remanences.

GA granites

Preliminary AF and thermal treatments on the GA granites revealed that all samples had at least two remanence components, and the directions of one hard (high coercivity or high T_{UB}) and two soft (low coercivity or low T_{UB}) remanences were calculated using orthogonal diagrams. Typical behaviour of three samples during AF treatment from three different sites is shown in Fig. 5. In the case of specimen GA 1-1.A at least two components with 30% of NRM intensity were removed, one below 20 mT, the second between 20 and 30 mT. The harder remanence was isolated above 30 mT, the straight decay to the origin indicating that no remanence of any stability was left in the specimen. The hard remanence (denoted as NW) of this specimen has a direction D = 325°, I = +20° and the soft components in order of hardness are SE and NE-1, respectively, characteristic of the GB granites.

In specimen GA 4-5D, a remanence removed between NRM and 40 mT with ≃30% of NRM intensity has a direction D = 260°, I = +65° similar but opposite in direction to the NE-1 remanence of the GB granites. The trajectory between 40 and 70 mT is interpreted as a hard magnetization (NW component) since little or no other remanence appears to remain and the decay is

approximately linear toward the origin. The behaviour of the hard component is similar to the one seen in specimen GA 1-1.A.

The AF demagnetization diagrams for specimen GA 7-5A show a remanence directed approximately along the PEF, with $D = 350°$, $I = +70°$ and an intensity of 20% of NRM. This remanence is considered to be of recent origin. Again, a NW remanence is removed between 40 and 80 mT with a typical linear trajectory towards the origin. Corresponding GA granite specimens during thermal demagnetization reveal the same two remanences with $T_{UB} \leq 350°C$ and $T_{UB} = 350–550°C$ but the AF treatment always appeared to separate them more effectively. In view of the presence and variable nature of different remanences at different sites, more than 70% of the collection (≈70 specimens) was treated up to 80 mT in 15 steps and the rest was demagnetized with a minimum of three steps.

In summary the most important feature of the GA granites is the presence of two remanences, which vary from sample to sample at any site, though the hard NW remanence always seems to reside in the higher coercivity part of the spectrum between 35 and 80 mT. The soft remanences apparently always correspond to the SE, NE-1, and NE-2 of GB granites (Table 3).

Statistical analysis of site-mean directions

Site-mean NRM directions for the GB and GA granites are not tabulated since the individual specimen directions show the major features, namely the wide scatter and complexity of NRM in the GB collection and the apparent influence of SE, NE-1, and NE-2 remanences in the GA collection (Fig. 3).

GB site-mean directions obtained after AF treatment between 10 and 60 mT are listed in Table 2. Whereas the sites 1 (group SE), 2 (group SE), 3 (group NE-1), and 7 (group SE) were found to possess single remanences, the other sites 4 (groups NE-2 and NE-1), 5 (groups NE-2 and NE-1), and 8 (groups NW and SE) were associated with both the low and high coercivity remanence directions. At site 6, we observed in three samples a NW direction along with a NE-2 component and in two samples a single NE-2 direction. The NE-1 directions obtained at sites 3 and 4 were inverted in conformance with the direction of opposite polarity in three GA sites (Table 3) and at one GB site 5 (Table 2) for statistical purposes (Table 5) but the NE-1 notation was retained for westerly directions. A similar direction ($D = 279°$, $I = +70°$) was also found in a contemporaneous New Brunswick rock unit with dual polarities (Roy *et al.* 1979). The corresponding site-mean directions (based on sample means, Table 2) of GB granites are shown in Fig. 6 and the corresponding group-mean directions based on sites are given in Table 5.

The NW remanence is present only at GB sites 6 and 8, as is the case with the GA sites. The SE, NE-1, and

TABLE 2. Site-mean directions of magnetization of GB granites after AF treatment

	SE					NE-1					NE-2					NW				
Site	$N(n)$	$D°$	$I°$	K	$\alpha_{95}°$	$N(n)$	$D°$	$I°$	K	$\alpha_{95}°$	$N(n)$	$D°$	$I°$	K	$\alpha_{95}°$	$N(n)$	$D°$	$I°$	K	$\alpha_{95}°$
1	5(12)	143	+69	323	4.3															
2	3(9)	149	+66	174	9.4															
3						5(14)	66	−70	337	4.2										
4						3(8)	64	−69	920	4.1	5(12)	22	−30	267	4.7					
5						4(4)	267	+69*	57	12.3	4(8)	32	−22	51	13.0					
6											5(10)	39	−18	310	3.8	3(5)	314	+12	51	17.5
7	3(11)	131	+74	1239	3.5															
8	3(9)	170	+69	1719	3.0											3(9)	322	+19	192	8.9

NOTES: N is number of samples used in statistics; n is number of specimens; $D°$, $I°$ are declination, inclination of remanent magnetization; K is estimate of Fisher's precision parameter; $\alpha_{95}°$ is semi-angle of cone of 95% confidence for mean direction.
*In computations of Table 5, the directions have been inverted for site 5 but the notation given to the direction, NE-1, is retained.

TABLE 3. Statistical summary of site-mean soft remanence directions at GA sites

Site	SE				NE-1				NE-2				NE-3*			
	$N(n)$	$D°$	$I°$	K	$\alpha_{95}°$	$N(n)$	$D°$	$I°$	K	$\alpha_{95}°$	$N(n)$	$D°$	$I°$	K	$\alpha_{95}°$	$N(n)$	$D°$	$I°$	K	$\alpha_{95}°$
1	4(11)	140	+71	120	8.4	2(4)	270	+76	—	—						1(2)	81	+73	—	—
2											4(5)	25	−21	120	8.4	1(2)	76	+60	—	—
3	3(4)	145	+64	62	15.9						3(5)	20	−10	110	11.9					
4	1(1)†	180	+75	—	—						5(9)	24	−11	98	7.8					
5	3(5)	164	+71	89	13.1											2(5)	94	+69	—	—
6						2(2)	279	+76	—	—	3(6)	66	−16	270	7.6					
7						5(6)	256	+74	300	4.3										
8	5(6)	116	+72	81	8.6															
9	3(9)	136	+73	76	14.3						2(5)	30	−17	—	—	2(5)	54	+74	—	—

NOTE: Symbols are as in Table 2.
*Since this is not a common remanence observed in the granites, the corresponding group-mean direction is not calculated (Table 5).
†Direction not used in group-mean statistics (Table 5) as this is obtained from $N(n) = 1$ only.

NE-2 remanences in the GB samples reside predominantly in the 10–60 mT coercivity spectrum. In GA samples the same components were commonly found to be confined to a narrow coercivity range of 5–25 mT (Table 3).

At GA sites 1, 2, 5, and 9 some samples revealed a northeasterly directed, steep downward direction (NE-3; $D = 77°$, $I = +69°$), not common in GB samples (Table 3). Finally, at the higher end of the coercive force spectrum (20–70 mT), a stable NW component is always found to be present (except at site 5, Table 4) with SE, NE-3, NE-2, and NE-1 remanences having lower coercivities (Table 3). Group-mean directions for the GB and GA sites are listed in Table 5.

Paleomagnetic poles and their ages of magnetization

The SE, NE-1, and NE-2 remanence directions obtained from the GB and GA granites do not differ significantly from each other (Table 5) except for differences in the range of coercivities with which they are associated (10–60 mT in GB, 5–25 mT in GA). Furthermore, in the GA collection the directions are always associated with a harder (20–70 mT) NW remanence (Fig. 5). Because of this similarity in the remanence group directions at GB and GA sites,

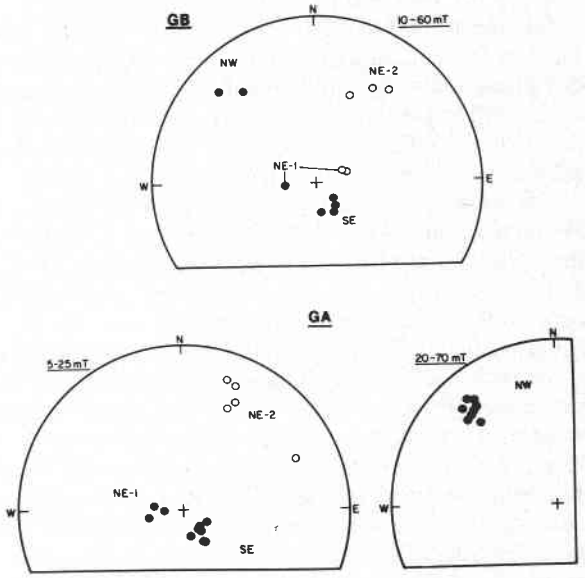


FIG. 6. Site-mean directions of magnetization plotted on Wulff net for GB (top) (Table 2) and GA (bottom) (Table 3) granites after AF treatment as indicated. The respective site-mean directions are identified groupwise based on the pattern of declinations: southeast (SE), northeast (NE), and northwest (NW). Note also that the group remanence NE-1 (northeasterly and steep upward inclination) identified for the two GB sites 3 and 4 (Table 2) is retained for the opposite polarity, westerly directions ($D \simeq 260°$, $I \simeq +70°$). The other symbols are as in Fig. 3.

TABLE 4. Site-mean directions of NW remanence observed at GA and GB sites, after AF treatment

Site*	$N(n)$	Treatment (mT)	$D°$	$I°$	K	$\alpha_{95}°$
GA-1	5(17)	30–75	314	+17	26	15.3
GA-2	4(6)	30–60	316	+12	36	15.6
GA-3	3(10)	25–60	321	+15	38	20.2
GA-4	4(10)	40–70	317	+16	102	9.2
GA-6	3(8)	30–60	320	+10	92	12.9
GA-7	5(8)	40–80	318	+22	37	12.7
GA-8	5(8)	25–75	319	+15	93	8.0
GA-9	1(1)	40–70	294	+14	—	—
GA-10	5(7)	30–80	322	+12	33	13.5
GB-6	3(5)	30–80	314	+12	51	17.5
GB-8	3(9)	30–80	322	+19	192	8.9
Total (10 sites)			318	+15	298	2.8

NOTE: Symbols as in Table 2.

*Two GA sites (5, no NW remanence is observed (Table 3); 9, direction obtained from only one sample) are not included in the group-mean statistics (also Table 5).

combined group-mean remanence directions based on site unit weight have been calculated and the corresponding paleomagnetic poles listed (Table 5).

Though paleomagnetic sampling was concentrated at or near radiometrically dated localities (Table 1), it is rather difficult to establish the sequence of acquisition of remanences (Table 5). Firstly, a given isotopic age does not necessarily correspond to a paleomagnetic pole from the same rock unit. Intrusives generally possess complex histories of remanent magnetization associated with several variables depending on their thermal history and postemplacement local and regional tectonics such as tilting and rotation (Pitcher 1979). Secondly, one cannot assess with certainty which of the remanences was acquired earlier simply from the coercivity or unblocking temperature spectra, since the remanence associated with the higher ends of the spectra is not necessarily the earlier one (Buchan and Dunlop 1976). Keeping these ambiguities in mind, possible best estimates of the ages were assigned to the corresponding paleomagnetic poles (Table 5) based on the available radiometric data along with the commonly observed magnetization characteristics and the published paleomagnetic data available from contemporary rock units obtained from New Brunswick and Newfoundland.

The results obtained in the present study can be assigned broadly to two widely different ages of magnetization based on the available isotopic data.

(1) The high coercive (20–70 mT) NW remanence that is found to be always associated with GA granites (central Cape Breton Island, Fig. 2), which have radiometric ages roughly between 520 and 580 Ma (Table 1), corresponds to pole CB (Table 5). An exception is site GA-5, characterized by SE and NE-3 remanences (Table 3).

(2) The other components SE, NE-1, and NE-2 (with poles CB-1, CB-2, and CB-3), which are often found in a broad coercivity spectrum (10–60 mT) in GB granites (northern Cape Breton Island, Fig. 2). Their ages range from 350 to 450 Ma (Table 1) and differ in age by at least 100 Ma from the GA sites. In the GA granites the S, NE-1, and NE-2 remanences are always present as low coercivity (5–25 mT) magnetizations along with another soft component NE-3, which is not common in the GB sites. This is consistent with observations by Wiebe (1972) that the older granitic rocks were affected by the Acadian Orogeny in Siluro-Devonian times. Exceptions to the above generalization are the GB sites 6 (coastal gneisses of Wiebe 1972) and 8 (Black Brook granites) where a NW remanence is present; the corresponding locations of the sites are close to locality 15 of Cormier (1972) (Table 1).

(3) The most commonly observed SE remanence is the only remanence found at the GB sites 1, 2, and 7 (Table 2), which form part of Cape Smoky and White Point plutons (Table 1). Their primary isotopic ages range from 410 to 460 Ma. The corresponding pole CB-1 compares well with a pole JC (Table 6, Fig. 7), observed in the approximately contemporaneous Mascarene Group intrusives (410–420 Ma), New Brunswick (Roy *et al.* 1979). At GB site 8 and GA sites 1, 3, 5, 8, and 9 (Tables 2, 3) the SE remanence coexists with the high coercivity NW remanence. Perhaps this means that the SE remanence was acquired some time later (up to 100 Ma) than the acquisition of the NW remanence.

(4) It is difficult to determine the sequence of acquisition of the NE-1 (pole CB-2) and NE-2 (pole CB-3) remanences since both often coexist at the same sites.

This is analogous to the results of Roy *et al.* (1979) who found two remanences in the St. George pluton yielding poles JA and JD (Table 6, Fig. 7). Likewise the

TABLE 5. Statistical summary of group-mean directions of magnetization and paleomagnetic poles

Group	GB granites[a]				GA granites[b]				GB and GA combined[c]				Paleopole				Inferred age of magnetization (Ma)
	N	$D°$	$I°$	K	$\alpha_{95}°$	N	$D°$	$I°$	K	$\alpha_{95}°$	N	$D°$	$I°$	K	$\alpha_{95}°$	Lat.°	Long.°	δp°	δm°	Pole[d]	
SE	4	149	+70	160	7.3	5	141	+71	149	6.3	9	145	+71	166	4.0	16N	41W	6	7	CB-1	430–460
NE-1	3[e]	72	−70	320	6.9	3	268	+76	650	4.8	6	256	+73	228	4.4	32N	97W	7	8	CB-2	400–430
NE-2	3	31	−24	67	15.1	5	33	−16	19	17.9	8	32	−19	27	10.8	27S	96W	6	11	CB-3	~400
NW	2	318	+16			8[f]	318	+15	320	3.1	10	318	+15	298	2.8	37N	176E	2	3	CB	520–580

NOTES: N is number of sampling sites used in statistics; δp°, δm° are semi-axes of ovals of 95% confidence; symbols are as in Table 2.
[a]Group-mean directions obtained from Table 2.
[b]Group-mean directions obtained from Table 3.
[c]Combined group-mean remanences based on Tables 2, 3.
[d]Antipoles for the corresponding paleopoles CB-1, CB-2, CB-3 are plotted in Fig. 7; paleopole CB is plotted in Fig. 9.
[e]The NE-1 directions are inverted to combine with the corresponding opposite polarity directions at GA sites.
[f]NW group remanence obtained for GA (=8) sites, Table 4.

age sequence could not be established. It is more interesting to note that the CB-2 pole is in good agreement with their JA pole and the CB-3 pole with their JD pole (Fig. 7), strongly indicating that widespread tectonic events are at the origin of their remanence acquisition. Of much interest to this problem is the result (pole MS, Table 6, Fig. 7) of Seguin *et al.* (1980) obtained from the Compton Formation, which belongs to the Gaspe – Connecticut Valley synclinorium in the Quebec Appalachians. The fossil plant age of this formation is Early Devonian and the pole was derived from a remanence with a positive fold test and a positive contact test. Pole MS is in reasonably good agreement with both poles CB-3 and JD. Furthermore, recently reported paleomagnetic results by Spariosu and Kent (1980) on Lower Devonian Traveler felsite from northern Maine (pole TF, Table 6, Fig. 7), by Brown (1980) on Devonian Dockendorff volcanics, also from northern Maine (pole DG, Table 6, Fig. 7), and by Roy and Anderson (1981) on upper Upper Silurian feldspathic graywackes from New Brunswick (pole JE, Table 6, Fig. 7) are also in good agreement with the pole CB-3, suggesting a remanence recording event possibly of ≤400 Ma age for these poles. Perhaps the CB-2 and JA poles are derived from an older remanence acquired during emplacement of the intrusives, which according to Cormier (1972) have ages ranging from 350 to 450 Ma in Cape Breton. The CB-2 and JA poles are also in rough agreement with the poles obtained from Silurian Mount Peyton diorite (≃420 Ma; pole MP-D, Table 6, Fig. 7) of Lapointe (1979) and from the Bonavista Bay dikes (between 350 and 400 Ma, pole BD-2, Table 6, Fig. 7; Murthy and Rao 1978).

(5) The NE-3 magnetization direction (D = 80°; I = +70°) is not as common as the other remanences in the GB sites but is present only in six samples at four GA sites (Table 3) along with the high coercivity NW remanence. Similar remanence directions were also common in the St. Stephen intrusive (*B* component: D = 89°, I = +82° with pole JB, Table 6) of New Brunswick (Roy *et al.* 1979) and in Mount Peyton granite (D = 1°, I = +83° with pole MPG, Table 6) of Newfoundland (Lapointe 1979), the average ages of which correspond, respectively, to 408 Ma (based on K–Ar age of hornblende: Wanless *et al.* 1973) and 380 Ma (based on RB–Sr age: Bell *et al.* 1977) but the later age (≃380 Ma) on the Mount Peyton granite is distinctly younger than the 420 Ma $^{40}Ar/^{39}Ar$ age reported recently by Reynolds and Morgan (1980). Since the NE-3 is not a common remanence direction, we did not calculate the corresponding pole.

A summary of the approximate ages of magnetization inferred in the present study as shown in Table 5 suggests that the poles CB-1 to CB-3 do not differ in their ages by more than a maximum of 50 Ma from each other, which may mean that the plutons were emplaced

during a short interval of 30–50 Ma. These magnetization ages appear closely related to Acadian deformation in Cape Breton Island (Wiebe 1972; Cormier 1972; Helmstaedt and Tella 1973). But the corresponding pole displacement is considerable in such a short time interval, which may in turn mean a rapid apparent polar movement with respect to the study area.

Discussion of the paleomagnetic results

Following the pre-Ordovician tectonic zonation of the Appalachians into (1) ancient continental margin of eastern North America, (2) Andean type margin of the Iapetus, and (3) Avaland microcontinent by Williams (1978*a*), published paleopoles of Silurian and Devonian age from the Appalachians are tabulated according to the tectonic history of the respective sampling areas (Table 6) and are shown in Fig. 7 along with the poles obtained in the present study (CB-1 to CB-3, Table 5).

Silurian data

Silurian data from various paleomagnetic studies are summarized in Table 6 and Fig. 7. Though the published geological ages differ for the Rose Hill Formation (Mid-Silurian), Weedon-Gould metavolcanics (Mid-Silurian), the Lawrenceton Formation (Early Silurian), and the Wigwam Formation (perhaps younger than Early Silurian) the corresponding paleopoles RH, WM, B-RH, and B-RS do not differ significantly from each other when considering the associated error estimates, but disagree with the Late Silurian BB pole from the Bloomsburg red beds. Some possible causes for the latter discrepancy are: (1) an error in the assumption that the age of the BB pole is similar to that of the four other poles; (2) rapid apparent polar wander during part of the Silurian period; or (3) the presence of undetected local relative rotations (French and Van der Voo 1979). The most important fact is that no distinctive pattern emerges from the five Silurian poles as to the geographic distribution of their sampling locations, and the corresponding tectonic setting. They have moderate southerly paleolatitudes (RH = 26°, WM = 16°, B-RH = 28°, B-RS = 15°, BB = 17°) corresponding to the remanence inclinations. An estimate regarding the age of the CB-3 paleopole can be made as latest Silurian to Early Devonian since the pole falls in the general area (longitudes 80–90°E, latitude about 30°N, Fig. 7) of the other recently reported Early Devonian poles. But an interpretation of this kind depends on the reality of grouping of Early Devonian poles in the general area referred to above.

Since this paper was submitted for publication, Kent and Opdyke (1980) have reported two paleomagnetic results (pole HE, from uppermost Silurian – Lower Devonian Hersey Formation; pole EP, from Lower Devonian Eastport Formation; Table 6, Fig. 7) from eastern Maine. These two poles are in agreement with the other two poles obtained from central Newfoundland (B-RS, B-RH) but are remote from the Bloomsburg BB pole (Fig. 7). Their interpretation is that central Newfoundland forms part of the "Acadia" displaced terrain during Siluro-Devonian time (see discussion below). We are not certain whether the agreement between the two sets of poles is valid since Lapointe (1979) assigned an Early Silurian age to the poles B-RH and B-RS. Furthermore, two recently reported Devonian results also obtained from Maine (Early Devonian pole, TF; Mid-Devonian pole DG in Table 6, Fig. 7) are significantly different from the poles HE and EP.

Devonian data

When all the Devonian poles obtained prior to 1979 from the Appalachians are plotted they show an apparently distinctive geographic distribution with a latitudinal discrepancy between the coastal New England – Canadian Maritime region and the now contiguous part of North America. This observation led Kent and Opdyke (1978) to take it as evidence of about 1500 km of relative tectonic displacement. Their scenario is that the coastal Canadian Maritime – New England region is a displaced terrain known as Acadia, which was not an integral part of cratonic North America until about Late Carboniferous, the relative motion having taken place predominantly along shear zones during a short interval of time (≃20 Ma) in the Carboniferous (Kent and Opdyke 1979; Fig. 8*a*). Their hypothesis of the possible presence of a shear zone is consistent with the extensive Carboniferous strike-slip faulting in the Canadian Maritimes (Webb 1969; Fig. 8*b*). Based on additional paleomagnetic data from folded Catskill red beds and on paleolatitudinal considerations, Van der Voo *et al.* (1979) have supported the possibility that after Late Devonian time Acadia underwent a sinistral strike-slip motion relative to cratonic North America along transcurrent shear zones. However, Roy and Morris (1981) have argued that the available Carboniferous data set for the east coast Carboniferous basin and the craton is not precise enough in terms of the contemporaneity of the ages of magnetization of the respective Carboniferous poles to detect the extent of the motions of Acadia relative to the craton during a short interval of time as proposed by Kent and Opdyke (1978, 1979).

In contrast to the pole positions used in Kent and Opdyke's hypothesis (1978, 1979), the poles obtained in the present study (CB-1, CB-2) and those from New Brunswick (JA, JB, JC: Roy *et al.* 1979) and Newfoundland (MP-D, MP-G: Lapointe 1979; BD-1, BD-2: Murthy and Rao 1978) strikingly fall further south. Lapointe (1979) and Roy *et al.* (1979) have interpreted their data in terms of rapid apparent polar wander during Siluro-Devonian times with respect to Newfoundland and New Brunswick rather than using them as evidence for relative motion between their study areas and the

TABLE 6. Paleomagnetic poles discussed in the present study (Figs. 7, 9)

Rock unit and location	Age*	Pole position Lat.°	Long.°	δp°	δm°	(α_{95}°)	Pole†	Reference
		Cratonic North America						
Tapeats red sandstone; northern and central Arizona	Early and Middle Cambrian	5N	158E	2	4		T	Elston and Bressler (1977)
Waynesboro Formation, dominantly red sandstone; Virginia, Maryland	Early to early Middle Cambrian	28N	165E	11	22		WB	Watts *et al.* (1980 *b*)
Rome Formation, dominantly red sandstone; Virginia, Tennessee	Early to early Middle Cambrian	38N	142E	6	12		R	Watts *et al.* (1980 *b*)
Bradore Formation, dominantly red sandstone; western Newfoundland	Early Cambrian	29N	167E	3	6		B	Rao and Deutsch (1976)
Rose Hill Formation, grey, red, and dolomitic sandstones; Maryland, northwest Virginia	Middle Silurian	19N	125E			(6)	RH	French and Van der Voo (1979)
Bloomsburg Formation, siltstone; Pennsylvania, Maryland, Virginia, West Virginia	Late Silurian	32N	102E	6	10		BB	Roy *et al.* (1967)
Clam Bank Group, sediments; western Newfoundland	Early Devonian	28N	146E	6	12		CG	Black (1964)
Raisin River dolomite; Ohio	Middle Devonian	24N	165E	9	11		RR	Martin (1975)
Columbus limestone; Ohio	Middle Devonian	45N	120E	2	3		CL	Martin (1975)
Delaware limestone; Ohio	Middle Devonian	48N	118E	2	4		DL	Martin (1975)
Catskill Formation, claystones, siltstones, and sandstones; New York	Middle to Late Devonian	47N	117E	2	5		CK-1	Kent and Opdyke (1978)
Catskill Formation, sandstones; Pennsylvania, Maryland, Virginia	Late Devonian	44N	124E			(5)	CK-2	Van der Voo *et al.* (1979)
		Andean margin of Iapetus						
Botwood Group, Lawrenceton Formation, rhyolite; Newfoundland	Early Silurian	13N	125E			(9)	B-RH	Lapointe (1979)
Botwood Group, Wigwam Formation, red sandstones; Newfoundland	Early Silurian	25N	113E			(14)	B-RS	Lapointe (1979)
Weedon–Gould metavolcanics; Internal domain, Quebec	Middle Silurian	29N	125E	9	16		WM	Seguin (1980)
Weedon–Gould St Francis metasediments; Quebec	Silurian–Devonian	24N	149E			(14)	WS	Seguin (1980)
Mount Peyton batholith, diorite; Newfoundland	~420 Ma	15S	68E			(10)	MP-D	Lapointe (1979)
Hersey Formation, red beds, diabase sills; eastern Maine	Latest Silurian to Early Devonian	20N	129E	5	8		HE	Kent and Opdyke (1980)
Eastport Formation, red beds, lavas; eastern Maine	Early Devonian	24N	114E	7	11		EP	Kent and Opdyke (1980)
Traveler felsite; northern Maine	Early Devonian	29N	82E	6	11		TF	Spariosu and Kent (1980)

TABLE 6 (*Concluded*)

Rock unit and location	Age*	Pole position Lat.°	Long.°	δp°	δm°	(α_{95}°)	Pole†	Reference
Compton Formation, metasediments; Quebec	Early Devonian	20N	79E	3	6		MS	Seguin *et al.* (1980)
Dockendorff Group, igneous rocks; Maine	Middle Devonian	24N	84E			(7)	DG	Brown (1979, 1980)
St Cecile – St Sebastien medium granite; Quebec	Middle Devonian	58N	147E	14	23		MG	Seguin *et al.* (1980)
St Cecile – St Sebastien fine aplitic granite; Quebec	Middle Devonian	39N	158E	7	13		MF	Seguin *et al.* (1980)
St Cecile – St Sebastien Compton hornfels; Quebec	Middle to Late Devonian	41N	165E	5	9		MH	Seguin *et al.* (1980)
Bonavista Bay alkali basaltic dikes; Newfoundland	400–344 Ma	2N	141E	—			BD-1	Murthy and Rao (1978)
Bonavista Bay alkali basaltic dikes; Newfoundland	400–344 Ma	54S	80E	—			BD-2	Murthy and Rao (1978)
Mount Peyton batholith, granite; Newfoundland	~380 Ma	63S	125E			(14)	MP-G	Lapointe (1979)
		Avalon zone						
Cabot Group, Signal Hill, Blackhead Formations, red sandstones; Newfoundland	Late Hadrynian	11N	122W	10	16		SH	Nairn *et al.* (1959)
Ratcliffe Brook Formation, red sandstones; New Brunswick	Early Cambrian	10N	124E	14	21		RB	Black (1964)
Wabana Group, red sandstones; Newfoundland	Early Ordovician	28N	168W	2	4		W	Deutsch and Rao (1970)
Mascarene Group, feldspathic graywackes; New Brunswick	~420 Ma	28N	85E	2	4		JE	Roy and Anderson (1981)
Mascarene Group, siltstones; New Brunswick	420–400 Ma	5S	87E	8	11		JF	Roy and Anderson (1981)
Mascarene Group, intrusives; New Brunswick	420–400 Ma	2S	122E	28	35		JC	Roy *et al.* (1979)
St Stephen pluton, gabbro; New Brunswick	~408 Ma	43S	136E	35	35		JB	Roy *et al.* (1979)
St George pluton, gabbro; New Brunswick	390 Ma	28S	85E	14	15		JA	Roy *et al.* (1979)
St George pluton, gabbro; New Brunswick	390 Ma	29N	93E	9	15		JD	Roy *et al.* (1979)
Perry Formation, lavas; New Brunswick	Late Devonian	26N	109E	10	17		PL	Black (1964)
Perry Formation, lavas; Maine	Late Devonian	24N	128E	—			PL-1	Philips and Heroy (1966)
Perry Formation, sediments; New Brunswick	Late Devonian	32N	118E			(~15)	PS	Robertson *et al.* (1968)
Perry Formation, sediments; New Brunswick	Late Devonian	35N	121E	5	9		PS-1	Black (1964)
Mafic complex; Massachussetts	Late Devonian	23N	126E			(10)	MC	Schutts *et al.* (1976)

NOTE: Symbols as in Tables 2, 5.
*Probable geological age of the rock unit, which may correspond to the time of magnetization as cited by respective authors.
†Pole symbols used in Figs. 7, 9.

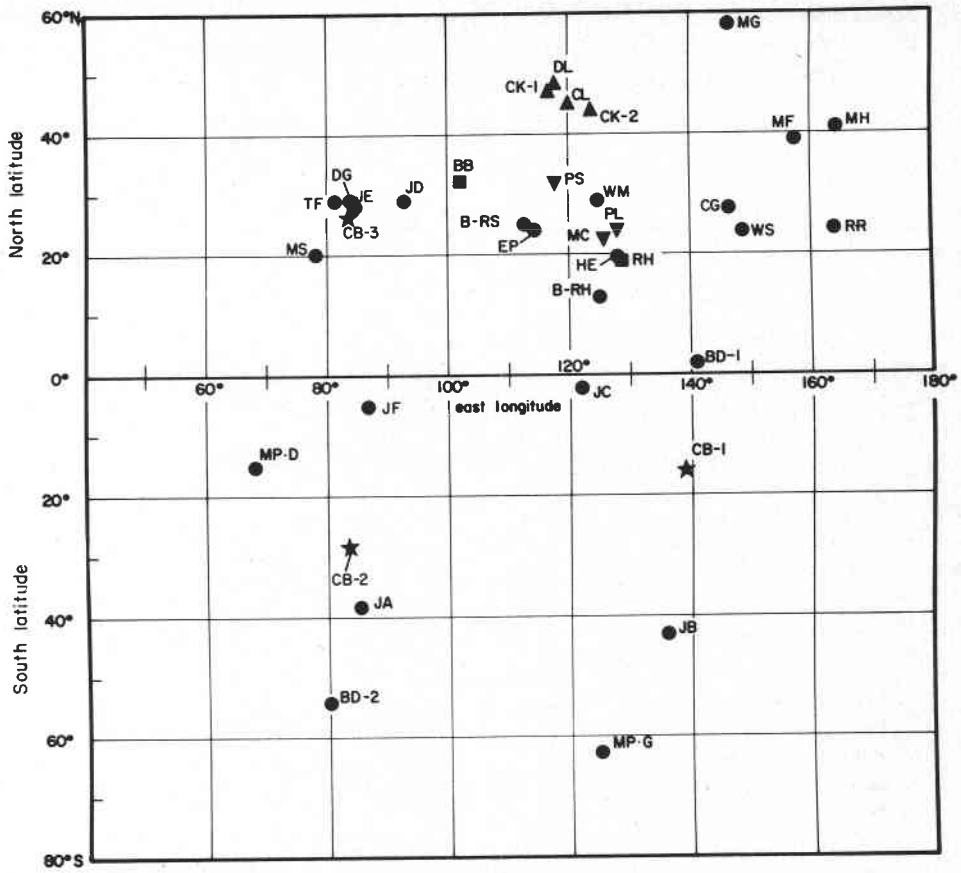


FIG. 7. Poles obtained from paleomagnetic studies of Silurian and Devonian rock units from Appalachians (Table 6). Poles quoted in abstracts are also employed. Two poles, one from Belchertown pluton (tectonic rotation; Ashwal and Hargraves 1977) and the other from Onondaga limestone (Permian magnetization; Kent 1979) are not plotted. For clarity, poles PL-1, PS-1 are not plotted. Pole symbols are as in Table 6. The solid squares represent Silurian poles (BB, RH) obtained from interior North America. The solid upright (inverted) triangles represent, respectively, the paleopoles derived from north (south) of Kent and Opdyke's (1978) hypothesized shear zone (Fig. 8*a*). The solid circles represent the other Devonian or Siluro-Devonian poles. The stars represent the poles obtained in the present study (Table 5).

North American craton, arguing that the paleomagnetic data base for the Silurian to Carboniferous interval is much too scanty to draw meaningful conclusions or formulate hypotheses such as that of Kent and Opdyke (1978).

Before discussing the significance and reliability of poles obtained from the Avalon zone, it should be pointed out that there are problems associated with the ages of magnetization of the Devonian poles from Laurentia and Acadia, which are critical to Kent and Opdyke's (1978) hypothesis. Whereas the Late Devonian pole CK-1 (Kent and Opdyke 1978) was obtained from a nearly flat-lying Catskill sequence, the age of pre-folding magnetization from the folded Catskill red beds for pole CK-2 has been inferred to be pre-Late Carboniferous (Van der Voo *et al.* 1979). Martin (1975) reported three Middle to early Late Devonian paleomagnetic poles from limestones and dolomites out of which poles DL and CL are close to poles CK-1 and CK-2; in contrast, another pole (RR, not used by Kent and Opdyke 1978) falls south of the other four poles and differs in longitude at least by 40° (Table 6, Fig. 7). Pole MC is based on a mean magnetization direction obtained from 22 oriented samples (rocks from gabbro to quartz diorite intrusives, mafic migmatites, and flows), the cleaned directions of which vary considerably from $D = 135–194°$ and $I = +4–+67°$ (Table 2 in Schutts *et al.* 1976). Pole PS is derived from the Perry sediments of New Brunswick, the age of which is considered to be either Late Devonian or Mississipian. According to the authors (Robertson *et al.* 1968), the sediments have a rather complex, unresolvable magnetic history and the suggested error is uncertain at least by 15°. Pole PL is from the Perry volcanics (Black 1964), and is based on

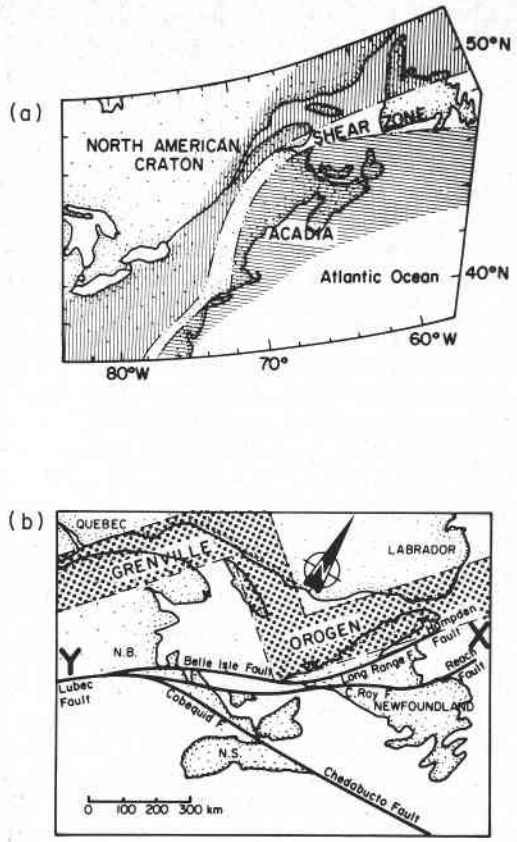


FIG. 8. (*a*) Map of eastern North America with approximate extent of mid-Paleozoic shear zone as marked between interior North America and Acadia following Kent and Opdyke (1978). (*b*) Simplified Late Devonian palinspastic restoration of Canadian Appalachians (Webb 1969) with mid-Paleozoic oblique continental collision model (Haworth 1975) between a laterally stepped Grenville continent (dark dots) and a straight-edged African continent (small dots) with the subsequent development of two major sets of transcurrent faults. The "aberrant" poles derived from sampling areas are located south of line XY.

AF demagnetization up to 30 mT. But Robertson *et al.* 1968) found the same Perry basalts to be magnetically unstable upon thermal demagnetization. In summary the data set used by Kent and Opdyke (1978) is not conclusive to prove their hypothesis.

"Aberrant" paleopoles

Paleomagnetic pole positions are calculated on the assumption of a geocentric axial dipole field averaged over a geologically short interval of time. The paleomagnetic argument for continental drift is based on the fact that clusters or groups of contemporaneous poles from different continents do not coincide with each other. If any paleopole of a given age falls more than a few tens of degrees outside such a cluster, then in the literature such a pole is usually referred to as "discordant" or "aberrant."

Such discordance could be attributed to one or more of the following causes. (1) The nature of the geomagnetic field is anomalous, for example if a nondipole component is dominant. (2) The rock unit under study is not magnetized parallel to the ambient field direction, for example due to strong anisotropy. (3) There is incomplete removal of stable or unstable magnetization components during demagnetization procedures, for example due to deep burial and subsequent uplift (Pullaiah *et al.* 1975), overprinted on the primary components. (4) The magnetization ages of different paleopoles being compared do not belong to the same age group because of uncertainties in dating the rocks. (5) There were postemplacement tilting, translation, and (or) rotation on a local or regional scale (MacDonald 1980). (6) The present plate to which the paleopoles refer includes one or more exotic fragments that became attached to the present plate during tectonic interactions of major plates (Wilson cycle). (7) A rapid apparent polar movement occurred during a short interval of geological time.

A cluster of Siluro-Devonian poles is found near longitude 120°E and between latitudes ~15 and 50°N (Fig. 7). Then according to the above definition all the poles, particularly those in the south latidudes, could be classified as aberrant. It is therefore worth investigating the cause of the aberrancy when the sense of divergence of the discordant poles is consistent. If the Siluro-Devonian cluster in northern latitudes is taken to represent the mid-Paleozoic field then cause (1) cannot be accepted. However, if the cluster is artificial (e.g., the result of overprints), then the anomalous nature of the geomagnetic field cannot be entirely discounted. It is highly unlikely that the discrepancy could be due to causes (2) and (3) as these are not applicable either to the present study or to studies by Lapointe (1979) and Roy *et al.* (1979). The age of the rock units (cause 4) studied here is well known and so are the rock units from New Brunswick and Newfoundland (Lapointe 1979; Roy *et al.* 1979) yielding the aberrant poles (Table 6, Fig. 7). Small undetectable postemplacement movements of intrusives are always a possibility (cause 5). However the agreement obtained from rock units sampled in Cape Breton (CB poles), New Brunswick (J poles), and Newfoundland (BD poles) indicates that large movements are most improbable. Indeed, the agreement between the pole clusters (CB-1, JC, BD-1; CB2, JA, BD-2) obtained from such widely separated areas is reasonably good and argues against appreciable local movements. Furthermore, geological tilting of several tens of degrees would be needed to make the aberrant poles coincident with either the Acadian or Laurentian poles. No geological evidence is present to account for

such local-scale tilting or rotation associated with the intrusives of the present study. It is also less likely on geological grounds that large granitic plutons would undergo large amounts of undetected tilt.

Then the only remaining possibilities for discrepancy could be due to causes (6) and (7). There have been repeated suggestions in the geological literature over the past two decades that a regional transcurrent motion (cause 6) has played an important role in the tectonic evolution of the northern Appalachians. Such a motion probably would be associated with a narrow belt of large strike-slip faults (Wilson 1962; Webb 1963, 1968, 1969), which may have been rejuvenated through the Acadian and Hercynian Orogenies.

Using regional geophysical data from Atlantic Canada, Haworth (1975) proposed a mid-Paleozoic oblique collision between a laterally stepped Grenville continent and a straight-edged African continent to show the development of the two major sets of transcurrent faults as defined previously by Webb (1969) (Fig. 8*b*). The two fault systems are, respectively, (1) the Lubec (New Brunswick), Belle Isle (Nova Scotia), Long Range, Hampden, Cape Ray, and Reach (Newfoundland) faults, which lie approximately parallel to the continental margin and (2) the Cobequid–Chedabucto fault system that lies oblique to the continental margin, identified mainly by the trend of the Collector magnetic anomaly. These two fault systems are associated with shear zones in mid-Paleozoic time. In a series of papers, Lefort and Haworth (1978), Haworth and Lefort (1979), and Haworth (1980) have demonstrated that three shear zones can be recognized based on geophysical data: (1) an extension of the south Armorican shear zone, which continued to north of Flemish Cap; (2) an eastward extension of the Cobequid–Chedabucto–Scatarie fault to northern Spain and southern France; and (3) the Clinton–Newbury fault of New England towards mid Spain.

Geological evidence is also consistent with such late Paleozoic strike-slip faulting in southern Europe and northern Africa (Arthaud and Matte 1977). Based on geochemical evidence Strong (1979*b*) suggested that most of the granitoid plutonism following the closure of Iapetus is related to crustal melting, shearing, and relative plate rotation along the "Hercynian megashear." While Williams (1979) finds it difficult to reconcile the movement of one part of an orogen relative to another in the compelling presence of the continuity of Ordovician and earlier tectonic elements along the full length of the Appalachians, he did not rule out such a possibility provided significant zones of sinistral strike-slip motion are present that parallel the tectonostratigraphic boundaries. Harland (1980) summarizes the problems of mid- to late Paleozoic continental reconstructions and the possibility of extended transcurrent motion between Mid-Ordovician and Mid-Carboniferous time. He favours a kind of transcurrent movement that was active during different geological times in different parts of Appalachians but mostly during the Carboniferous.

Based on paleomagnetic data for the first time, Morris (1976) proposed the existence of transcurrent motions in the Appalachian–Caledonide Orogen during pre-Late Devonian time. Piper (1979) interpreted the Caledonian paleomagnetic data in terms of a major strike-slip motion ($\simeq$ 3600 km) along or close to the line of orthotectonic Caledonides prior to Mid-Devonian times. Van der Voo (1980) considers the paleomagnetism of peri-Atlantic continents to favour a sinistral displacement between Late Devonian and latest Carboniferous time. Irving (1979) and Irving and Morel (1980) proposed that displaced terrains of early and mid-Paleozoic age may be present in the Appalachians and that these displacements could have occurred as part of a Wilson cycle.

All the above aberrant poles have high or intermediate paleolatitudes, which may mean that Acadia was situated very much south of its present position relative to North America during Siluro-Devonian times, as maintained by Morris (1976) for pre-Late Devonian and Kent and Opdyke (1979) and Van der Voo (1980) for pre-Late Carboniferous times. As a possible explanation for intermediate to high paleolatitudes, Irving (1979) suggested that Acadia may have been a sliver of continental crust that was formerly part of northwest Gondwana and became attached to Laurasia during Siluro-Devonian time.

From the geological arguments mentioned earlier, it is difficult paleomagnetically to dispute the possibility of pre-Late Devonian transcurrent motion to bring the Acadia displaced terrane from southerly paleolatitudes to its present position. For example, in the Devonian palinspastic restoration map of Webb (1969) (Fig. 8*b*), all the sampling localities from which the aberrant poles were derived lie south of the Reach and Cape Ray faults in Newfoundland, of Cape Breton Island between the Reach and Chedabucto faults, and of the Lubec fault in New Brunswick, implying that the shear zone in fact was further south (approximately south of line XY, Fig. 8*b*) during pre-Late Devonian time, in contrast to the location of Kent and Opdyke (1978)'s hypothesized shear zone (Fig. 8*a*). For example, the two poles CB-2 and CB-3 may be interpreted as indicative of two different positions of Cape Breton before (southerly) and after (northerly) it adjoined cratonic North America during Siluro-Devonian time. One possible consequence of a zone of distributed shear in time and location (as proposed by Harland 1980) is the translation and rotation of rigid crustal blocks caught up within the shear zone in a ball-bearing fashion (Teissere and Beck 1973;

Beck 1976). But such postulated rotations cannot be easily accommodated and do not seem realistic since some of the "non-aberrant" poles such as B-RH and B-RS (obtained from Botwood Group rhyolite and red sandstone) would have suffered a rotation in the same fashion (clockwise or anticlockwise) as the aberrant MP-D and MP-G poles (obtained from the Mount Peyton batholith, which cuts the Botwood Group) and should have become aberrant by themselves. At the same time the concept of distributed shear in different tectonostratigraphic zones at different geological times prevalent in the northern Appalachians and their influence on the magnetization directions cannot be entirely discounted.

From the above arguments, the most viable explanation of the results would seem to be alternative (7), that is, rapid apparent polar wander relative to zones 2 and 3 (Table 6) during the time of emplacement of granitic intrusives; this is also the cause favoured by Roy *et al.* (1979). It remains to be seen whether the apparent polar wander path is valid for the whole of North America or not. At present the paleomagnetic data (Siluro-Devonian poles) plotted in Fig. 7 suggest the following features: (1) a cluster of Silurian poles that lie roughly between latitudes 15 and 30°N and longitudes 100 and 130°E; (2) a grouping of poles predominantly Early Devonian in age derived from zones 2 (Andean type margin of Iapetus) and 3 (Avalon zone) (Table 6) located approximately at 30°N with longitudes 80–90°E; (3) a concentration of poles, mostly Mid- to Upper Devonian in age, occurring near longitude 120°E and latitudes 20–50°N; (4) a poorly defined cluster of (aberrant) poles in the same longitude range (70–140°E) but occurring in southerly latitudes of (a) <20°S (MP-D, JF, JC, CB-1) or (b) 30–60°S (CB-2, JA, JB, BD-2, and MP-G). All these aberrant poles are mostly derived from intrusives with radiometric ages ≥400 Ma and from the eastern margin of the hypothetical Proto-Atlantic, but these results do not prove or disprove the geological integrity of North America during this time period. More well-dated and paleomagnetically reliable data from both interior North America and the Atlantic provinces are needed before hypotheses can be put forward.

Paleomagnetism of NW remanence

The high coercivity (35–80 mT) NW remanence is predominantly present in the majority of GA sites and in two GB sites (6, 8). The primary radiometric ages of these sites (Table 1) are in the range 520–580 Ma with a corresponding paleopole CB (Table 5), which differs in age from the other poles CB-1 to CB-3 by at least 100 Ma. It is believed that this residual NW remanence is primary because of its high coercivity and the absence of a major post-Avalonian tectonic event other than the Acadian Orogeny of Siluro-Devonian age. Other individual paleopoles derived from the Avalon zone are unfortunately not completely reliable and do not correspond to the age of CB (520–580 Ma). For example pole SH is based on NRM data; RB is based on limited demagnetization treatment (30 mT); W is based on yet unsolved complex thermal behaviour (Table 6, Fig. 9) and these cannot be compared with the CB to infer the paleogeography of the Avalon zone. In the absence of reliable paleomagnetic data from contemporaneous rock units in the Avalon zone, an attempt is made following Wilson's (1966) hypothesis to compare the CB pole to approximately contemporaneous poles derived from the early Paleozoic European and African fragments, which were contiguous and formed the eastern margin of the hypothetical proto-Atlantic by making a correction for the Mesozoic and Tertiary opening of the present Atlantic (Bullard *et al.* 1965). Since no reliable data for the age span 520–580 Ma are available from the Anti-Atlas region of northwestern Africa (Daly and Pozzi 1977; Morris and Carmichael 1978; Martin *et al.* 1978) and Great Britain (Duff 1980), comparison is made with reliable poles (St Quay diorite, 583 ± 40 Ma, SQ_1; microgranite dikes and contacts, 573–510 Ma, MI) from the Armorican massif, France (Hagstrum *et al.* 1980; Van der Voo *et al.* 1980). The corresponding relocated poles (35N, 101E; 15S, 92E) disagree with the CB pole. This may mean either the pivot location chosen for the reconstruction (Bullard *et al.* 1965) is in error or that the magnetization directions do not belong to the same age group. But such speculations on possible continental reconstructions for the period 520–580 Ma, at present based on one single result from the Avalon zone (North America), are premature. Furthermore, the apparent scatter and complexity of the Cambrian paleomagnetic poles from the cratonic margin of North

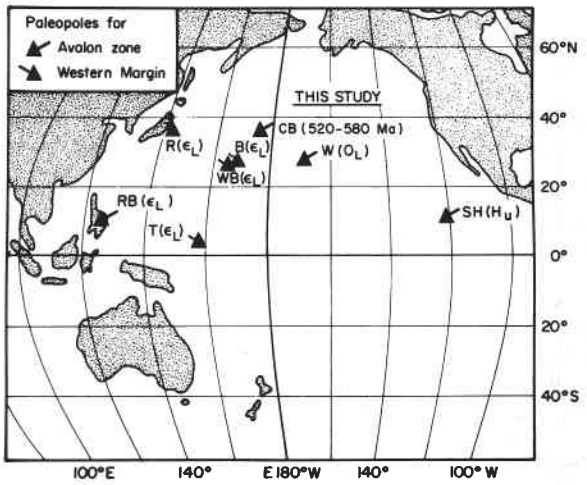


FIG. 9. Paleomagnetic poles (late Hadrynian, H_u; Early Cambrian, ϵ_L; Early Ordovician, O_L) derived from Avalon zone and cratonic North America (Table 6).

America (Watts *et al.* 1980*a*,*b*) preclude also the comparison of the CB pole with similar age results. The discordance of the CB pole from Early to early Middle Cambrian poles (T, B, WB, R) derived from the North American platform (Fig. 9) cannot be taken with certainty as an evidence for the existence of a proto-Atlantic ocean in early Paleozoic time.

Conclusions

Judging from magnetization data there are two generations of granites that differ in age at least by 100 Ma: the older Cambrian granites in the central part and the younger Siluro-Devonian intrusives in the northern part of Cape Breton Island. The younger granitic intrusives are characterized by three different ages of magnetization, the radiometric ages of which differ from each other by no more than 50 Ma, with an age range of 350–450 Ma; they correlate very well with the Siluro-Devonian Acadian Orogeny postulated for Cape Breton Island. Antipole CB-3 agrees with other recently reported Early Devonian paleomagnetic results (MS, TF, JE, JD, DG; Fig. 7) from the northern Appalachians. The two other antipoles, CB-1 and CB-2, are located in southerly latitudes and are not significantly different from contemporaneous poles derived from New Brunswick and Newfoundland. Based on the so-called aberrant paleopole data, the possibility of a pre-Late Devonian transcurrent motion in the Canadian Appalachians cannot be entirely discounted but because of the scanty data base it is far from being proven.

The older Cambrian granites give a stable high coercivity direction with a corresponding pole CB (37N, 176E), which probably represents the age of the Avalonian Orogeny (520–580 Ma). The older granitic gneisses in the north (Wiebe 1972) and the Cambrian granites in the central part of Cape Breton Island are characterized by secondary remanences of Siluro-Devonian age (SE, NE-1, NE-2), which is consistent with the geological evidence that the older plutons were affected during the Acadian Orogeny. In the absence of other contemporaneous poles from the Avalon zone, the single CB pole is inadequate to interpret the tectonics and paleogeographic setting of the Avalon zone.

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